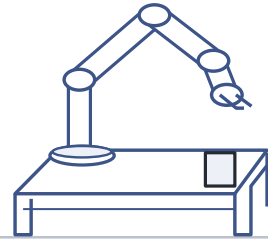


Source organization

Successful simulation replays

Expanded
in Fig. 2

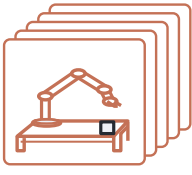
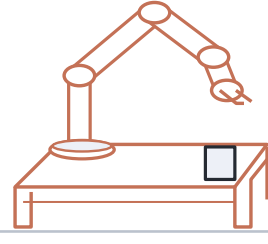
Successful replays (simulation)



- 50 episodes per task
- Current states actions A_{ref}

State records

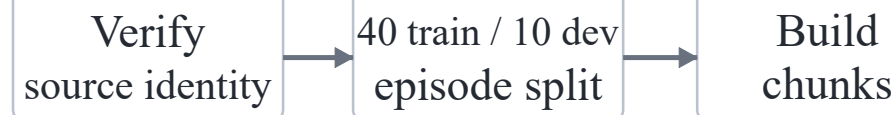
Reference actions (same episodes)



- 400 steps per episode
- 14 action channels actions A_{ref}

Action records

Preprocessing (build training records)



Training record
(same task)

A_{ref}
Reference actions

S_0
(initial layout for replay)

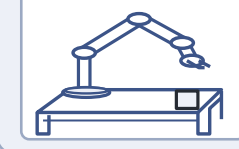
Policy-error calibration

Calibrate within each task

Expanded
in Fig. 3

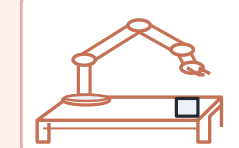
Episode-separated fitting (five auxiliary policies)

Excluded fold: 8



o_t
 Y_t
...

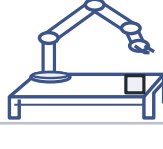
Fitting folds: 32



o_t
 Y_t
...

5-fold
episode
split
(per task)

Fitting supervision



o_t
 Y_t

Fit on 32
(o_t, Y_t)

$$L = \text{MSE}(f(S_o(o)), S_a(Y))$$

MSE loss
(fit folds only)

Out-of-episode calibration

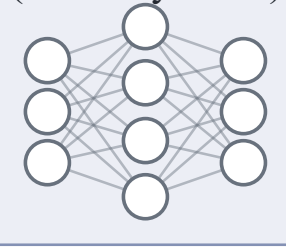
Current inputs: excluded fold
Policy fitted on the other 32

$q_t \cdot 14\text{-D}$
(robot state)

$p_t \cdot 3/6\text{-D}$
(object position)

$t/T \cdot 1\text{-D}$
(elapsed time)

$f\text{-fold}$
(auxiliary MLP)



$f = 1$
 $f = 2$
...
 $f = 5$

Pool 5 folds by time
1960 error chunks / task

Fit-fold optimization only

$$E_t = \hat{A}_t - Y_t$$

(raw action units)

Errors
 $\{E_t\}$

A_{ref} (reference sequence for replay)

s_0 (reset the same initial layout for each attempt)

Recovery supervision

Replay and record

Expanded
in Fig. 4

Bounded perturbation (8-step pulse)

$g = 0$

Protect grip

RMS scale

Cap $\pm 3\sigma$

$E_{1:8}$

A_{ref}
(ref.)

+ Add pulse

Replay

Replay audit (all trials)

A_{exec}
executed
source

$\tilde{o}_t$
observed
endpoint

r_{399}
terminal reward
seed / scale

Recovery windows $t \geq u$

Pair replay state with executed action

Recovery record (window)

Replay state $\tilde{o}_t$ (current)

$\oplus$ p $\otimes$ e $\uparrow$ q ⌚ t/T

$Y_t = A_{exec, t:t+16}$
(executed reference tail)

source · endpoint · start · seed · scale

Action-chunk policy
1:1 original + recovery